



Frontier Homeopathic Medical College

Digital Management System Project Overview Report

System Capabilities | Features | Future Roadmap

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Hayatabad, Peshawar, KPK, Pakistan | 2025

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Table of Contents

1.	Executive Summary	3
2.	About the Institution	3
3.	What Is This System	3
4.	System Modules and Features	4
5.	Management Portals	5
6.	AI Assistant Integration	5
7.	Current Delivery Status	6
8.	Future Development Roadmap	7
9.	Why This System Matters for FHMC	8
10.	Technical Foundation	8
11.	Contact and Next Steps	9

1. Executive Summary

This report presents a full overview of the Frontier Homeopathic Medical College (FHMC) Digital Management System, a purpose-built institutional platform developed specifically for FHMC, Hayatabad, Peshawar. The system has been designed to modernize the college's operations, strengthen its digital identity, and provide a seamless experience for students, faculty, and administration.

The platform is built on modern, enterprise-grade technology and is capable of supporting over 5,000 daily active users. It brings together a professional public website, a secure multi-role management portal, an AI-powered admissions assistant, and a real-time administrative dashboard into one unified system.

System Name	FHMC Digital Management System
Institution	Frontier Homeopathic Medical College (FHMC)
Location	Hayatabad, Peshawar, KPK, Pakistan
Developed by	Mubashir Noor mmubashir342@gmail.com
Technology	React 19, TypeScript, Vite, Supabase, Tailwind CSS
Daily Capacity	5,000+ Active Users (Scalable to 50,000+)
Current Status	Fully Functional System, Ready for Production Deployment

2. About the Institution

Frontier Homeopathic Medical College was established in 2015 under the leadership of its founder, Col. (Retd.) Iqbal Shaheen, and operates under the recognition of Khyber Medical University (KMU) and the National Council for Homeopathy (NCH). The college offers the five-year DHMS (Doctor of Homeopathic Medicine and Surgery) program, with practical clinical training beginning from the third year through in-house OPD facilities.

3. What Is This System

The FHMC Digital Management System is not simply a website. It is a complete institutional platform that combines a public-facing web portal with a fully secured internal management system. The two sides work together seamlessly:

Public Portal	Internal Management System
College homepage with hero slideshow	Admin dashboard with real-time controls
Programs and curriculum information	Teacher portal for attendance and scheduling
Online admissions application form	Student portal for fees, timetable, and records
Faculty and leadership profiles	Secure role-based login for all staff and students
Photo gallery and campus showcase	AI assistant trained on FHMC institutional data

Contact and location information

Live application tracking and management

4. System Modules and Features

The system is organized into distinct modules, each designed to serve a specific operational need of the college. Every module has been built with the requirements of a real academic institution in mind.

Homepage and Public Website

A visually impressive digital front door for the college. Features an animated photo slideshow of the campus, quick access to programs and admissions, and institutional highlights. Designed to create an immediate sense of professionalism and trust for prospective students and their families.

Online Admissions Module

A structured four-step online application system that guides applicants through eligibility verification, document submission, and application confirmation. Includes Pakistani-specific field formats, a mandatory KMC CAT entrance test reminder, and full eligibility criteria for F.Sc and DHMS pathways.

Faculty and Leadership Module

Official profiles of the Director (Col. Retd. Iqbal Shaheen), Principal (Dr. Fakhar), and Head of Department, with professional portrait photos. Communicates institutional credibility and academic leadership.

Gallery and Campus Module

A curated visual showcase of the campus, facilities, and college events with an expandable view to display the complete image collection without overwhelming the page layout.

AI-Powered Admissions Assistant

A built-in chatbot available across all pages that instantly answers questions about admissions, programs, eligibility, documents, and campus information at any hour of the day.

5. Management Portals

Upon secure login, the system automatically routes each user to their designated portal. All data and controls are strictly separated by role, ensuring privacy and security across the institution.

Administration Portal

- Review and manage all incoming admission applications with live status tracking
- Add, edit, and manage student and staff accounts across the system
- Oversee class schedules, academic sessions, and the institutional calendar
- Monitor financial records, fee collection status, and outstanding dues
- Generate reports and access institution-wide performance summaries

Teacher Portal

- Mark and manage student attendance for each subject and session
- Add new class sessions and update the academic schedule
- View the complete academic calendar with exam dates and events
- Access and update student academic performance records
- Communicate important notices to assigned student groups

Student Portal

- View personal attendance record and current percentage per subject
- Check fee payment history, due amounts, and payment deadlines
- Access full class timetable and upcoming examination schedule
- Track the status of submitted admission application in real-time
- Download academic documents and course-related materials

6. AI Assistant Integration

The system includes a purpose-trained AI assistant embedded directly into the website. It operates as a 24-hour admissions and information desk, serving prospective students and visitors without requiring any manual response from college staff.

Topic	What the Assistant Handles
Admissions Eligibility	F.Sc 50% rule, DHMS pathway, age and board requirements
Required Documents	Complete 7-item document checklist with details
KMU CAT Entrance Test	Test requirement, registration, and timing guidance
DHMS Program Information	Duration, year-wise curriculum, and clinical training structure
Campus and Location	Hayatabad address, directions, and office hours
Institutional Background	History, founder, KMU and NCH affiliation details

General Inquiries	Handles greetings and redirects off-topic questions professionally
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7. Current Delivery Status

The following table reflects the current build and delivery status of all system components.

Public Website with 7 Complete Pages	DELIVERED
Animated Hero Slideshow with Campus Photos	DELIVERED
Real FHMC Institutional Photos Integrated	DELIVERED
Director, Principal and HOD Profiles	DELIVERED
FHMC Official Branding, Logo and Favicon	DELIVERED
Online Admissions Form with 4-Step Process	DELIVERED
Admin Management Portal	DELIVERED
Teacher Attendance and Schedule Portal	DELIVERED
Student Records and Fee Tracking Portal	DELIVERED
AI Admissions Assistant (Fully Trained)	DELIVERED
Role-Based Secure Login System	DELIVERED
Pakistani Localization (Names, Context)	DELIVERED
Live Database Connection Setup	PENDING
Production Domain (fhmc.edu.pk or .com)	PENDING

8. Future Development Roadmap

The system has been architected for long-term growth. The following phases represent the planned expansion of the platform as FHMC's digital operations mature and scale.

Phase 1 : Live Production Launch

Connect the system to the live database, configure the official college domain, and make the platform fully accessible to students and staff. Real admission applications will begin flowing into the admin dashboard from day one.

Phase 2 : Academic Operations Management

Activate real-time attendance tracking by teachers, subject-wise grade entry and report card generation, automatic calculation of student attendance percentages, and end-of-term academic performance summaries accessible to both students and administration.

Phase 3 : Fee and Finance Management

Implement a structured digital fee management system with semester-wise payment schedules, automated outstanding due tracking, fee receipts downloadable by students, and a consolidated financial report dashboard for the administration.

Phase 4 : Document and Record Management

Enable students to upload admission documents directly through the portal, allow administrative staff to review, approve, or return documents with remarks, and maintain a digital archive of all student files for audit and compliance purposes.

Phase 5 : Communication and Notification System

Introduce automated email and SMS alerts for examination schedules, fee reminders, admission results, and academic announcements. All notifications are triggered automatically by system events without manual staff effort.

Phase 6 : Examination Management

Build a complete examination module covering timetable creation, seating plan generation, marks entry by subject teachers, automatic result compilation, and official result cards accessible to students through their portal login.

9. Why This System Matters for FHMC

A Single System for the Entire Institution

Managing admissions, attendance, fee records, examination results, and communication through separate tools or paper-based processes creates delays, errors, and administrative burden. This system brings every operation under one roof, accessible from any device at any time.

Admissions That Work Around the Clock

With the online admissions module and the AI assistant, prospective students can apply, get their questions answered, and track their application status at midnight or on a public holiday. The college never misses an inquiry again.

Real Accountability in Attendance and Fees

When attendance is marked digitally, it cannot be disputed or misplaced. When fee records are managed through the system, both the student and the administration have a clear, accurate, and permanent record at all times.

A Professional Identity That Commands Respect

In today's environment, a college's digital presence is often the first impression it makes. A polished, well-designed system communicates institutional credibility to students, parents, regulators, and peer institutions before a single meeting takes place.

Scalable Without Rebuilding

The system is designed to grow with FHMC. Whether the college adds a new campus, expands enrollment, or introduces new programs, the platform accommodates growth without requiring a new system to be built from scratch.

Reduced Administrative Workload

Tasks that currently require hours of manual effort such as compiling attendance sheets, sending fee reminders, or handling admission inquiries are automated by the system. Staff can focus on quality education rather than paperwork.

10. Technical Foundation

The system is built on technologies trusted by universities, hospitals, and enterprises globally. The following table summarizes the core technical components.

Layer	Technology	Why It Was Chosen
User Interface	React 19 + Vite	Industry-standard, fast, and highly maintainable
Code Language	TypeScript	Prevents errors, easier to scale and maintain
Visual Design	Tailwind CSS	Consistent, professional, and responsive design
Database and Auth	Supabase (PostgreSQL)	Enterprise-grade relational database with built-in security
AI Assistant	Custom Knowledge Engine	Trained on FHMC data, no external API dependency

Deployment	Vercel / Netlify CDN	Global availability, automatic updates, SSL secured
Version Control	Git and GitHub	Full code history, rollback capability, team collaboration

The system follows a three-tier architecture: a React-based frontend served via global CDN, a Supabase backend handling all data and authentication, and a custom AI layer for the assistant. This separation ensures each component can be maintained or upgraded independently without disrupting the rest of the system.

11. Contact and Next Steps

For a live walkthrough of the system, discussion of development phases, or any questions about the platform's capabilities, please reach out using the details below.

Developer	Mubashir Noor
Email	mmubashir342@gmail.com
LinkedIn	linkedin.com/in/mubashir-noor-mno
Location	Peshawar, KPK, Pakistan
Expertise	Events Operations, Digital Production, Web Systems and Institutional Platforms
Project	FHMC Digital Management System, Hayatabad, Peshawar

The FHMC Digital Management System is ready. The foundation is complete. The next step is bringing it fully live for the college and its community.